# 4.2 Reflections

Name _____ Period ____ Date _____

Geometry 4.2 (30-minute plan) · Student sheet

**I can…**

- reflect a figure in the x-axis, the y-axis, or a horizontal or vertical line.
- reflect a figure in y = x or y = −x.
- perform a glide reflection and find lines of symmetry.

**Vocabulary** · fill in the missing word

**Reflection** — a _____ that uses a line like a mirror to reflect a figure

**Line of reflection** — the line that acts as the _____

**Glide reflection** — a _____ followed by a reflection in a line parallel to the direction of the translation

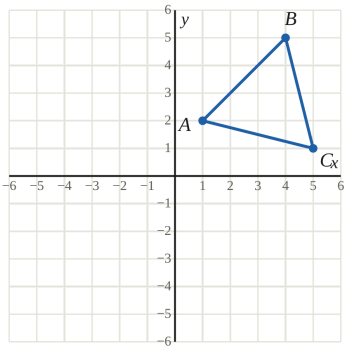
**Line symmetry** — a figure has line symmetry when a _____ maps the figure onto itself

## Part 1 · Notes and examples (fill in each note, then work the example)

**Reflections in the axes:** in the x-axis, (x, y) → _____; in the y-axis, (x, y) → _____.

### Example 1

Reflect △ABC in the x-axis, then (separately) in the y-axis. Give the vertices.




**Reflections in y = x and y = −x:** in y = x, (x, y) → _____; in y = −x, (x, y) → _____.

### Example 2

Reflect (3, −4) in the line y = x, then (separately) in the line y = −x.

**Glide reflection:** a translation followed by a _____ in a line parallel to the direction of the translation.

**Line symmetry:** a figure has line symmetry when a _____ maps the figure onto itself.

### Example 3

How many lines of symmetry does each figure have?

a. a rectangle that is not a square

b. an equilateral triangle

c. a regular hexagon

## Part 2 · Partner practice

### Partner practice

Reflect (6, 1) in the line y = x.